Course Name: Data Visualization

Course Outcome

CO1- Have a solid understanding of the data analysis process and the significance of EDA in data science.

CO2- Be proficient in using Python to perform data visualization and exploratory analysis.

CO3- Gain hands-on experience with popular data visualization libraries like Matplotlib, Seaborn, and Plotly.

CO4- Be equipped with the skills to create interactive dashboard using plotly and geospatial data analysis.

CO5- Be able to communicate complex data insights effectively through compelling data Visualizations

Printed Pages: **University Roll No. .....**

Mid Term Examination, Odd Semester 2023-24
B. Tech. (Hons.) CS, Year: II, Semester: III
BCSS 0051, Data Visualization

Time: 1 Hour **Maximum Marks: 15**

Section – A

*Attempt **All** Questions* 2 X 3 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Write a python code to plot a scatter plot, including title, xlabel and ylabel. Assume that the input variable is "years of experience", the output variable is salary and the dataframe is salary_data.	2	CO2	E	P
2	What is an interactive dashboard? Plot a histogram for the continent by using plotly, assume that the data frame is 'gapminder'.	2	CO4	An	C
3	Make a list of data analysis process steps.	2	CO1	U	F

Section – B

*Attempt **All** Questions* 3 X 3 = 9 Marks

No.	Detail of Question	Marks	CO	BL	KL
4	Consider the following Dataframe "data" which has the following columns: Age=[Nan,18,20,35,60,250,27,40] Salary= [2000,30000,40000,50000, Nan,37000,45000,34000] Gender=['M','M','F','F','M','M','F','F'] Purchase= [0,0,1,1,1,0,1,1] Note: output variable is "Purchase". By using the above data frame 'data' answer the following questions. Write a python code,	3	CO1	A & E	C

	i. To perform simple EDA and analyze how all the input parameters affect the outcome. ii. Use appropriate methods to replace Nan values. iii. Apply proper data engineering techniques to normalize the data. iv. Apply appropriate data engineering steps to convert categorical data to numerical data. Write your conclusion.				
5	How to perform Bivariate analysis for Numerical-numerical, Categorical-Categorical, and Numerical-Categorical variables?	3	CO2	E & C	M
6	Answer the following: i. What is the time series? ii. Which are the four important components which affect time series analysis? iii. What is an ADF test and why is it required?	3	CO3	U	C

Course Name: Probability and Applied Statistics

Course Outcome

CO1-List different kind of statistics for data analysis.
CO2-Differentiate measure of central tendency and measure of variability.
CO3-Define Normalization and Standardization.
CO4-Conceptualize Probability distribution.
CO5-Apply statistics in various research methods.

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Mid Term Examination, Odd Semester 2023-24
B. TECH(H), Year II, Semester III
BCSS0052: Probability and Applied Statistics

Time: 1 Hour **Maximum Marks: 15**

Section – A

Attempt All Questions 2 X 3 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	How is statistics important for data science?	2	CO1	U	F
2	What is the 68-95-99.7 rule for normal distributions explain how it can be used to answer questions about frequencies of data values in a normal distribution?	2	CO3	E	P
3	Consider a case in which analyzing the data from a survey, where people were asked questions on their favorite pets. Such dataset records the information from 50 respondents and the data is categorized into two variables: "Pet Type" and "Number of Pets Owned." "Pet Type" is a categorical variable that can have values like "Dog," "Cat," "Fish," and "Bird." "Number of Pets Owned" is a numerical variable representing how many pets each respondent owns. (i) Explain the difference between categorical and numerical variables using the given dataset as an example. (ii) Which variable, "Pet Type" or "Number of Pets Owned," would be suitable for creating a bar chart, and why?	2	CO5	A	C

Section – B

*Attempt **All** Questions* 3 X 3 = 9 Marks

No	Detail of Question	Marks	CO	BL	KL
4	Consider the following dataset representing the scores of 20 students in a mathematics test: 78,85,92,63,74,91,88,72,76,84,79,87,70,95,67,81,83,77,90,68. Calculate and draw a box-and-whisker plot for this dataset. Then, answer the following questions based on the plot: 1. What is the median (Q2) of the dataset? 2. What is the lower quartile (Q1)? 3. What is the upper quartile (Q3)? 4. What is the range of the data? 5. Are there any outliers in the dataset based on the box-and-whisker plot? Please provide your answers with calculations if applicable.	3	CO2	E	C
5	The probability is 1% that an electrical connector that is kept dry fails during the warranty period of a portable connector. If the connector is ever wet, the probability of a failure during the warranty period is 5%. If 90% of the connectors are kept dry and 10% are wet, what proportion of connectors fail during the warranty period?	3	CO3	C	M
6	Consider a case that you have a classroom of 30 students, and you want to understand their favorite colors. How would you use basic statistics to collect and summarize this information? Please outline the steps you would take, from data collection to presenting the results.	3	CO1	An	F

Course Name: ADVANCED DATABASE MANAGEMENT SYSTEM

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data models used in database design.
CO3: Design conceptual models of a database using ER modeling for real-life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys, using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss the recovery system and be familiar with the introduction to the web database, distributed databases.
CO8: Design and implement the database system with the fundamental concepts of DBMS using Python.

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Mid Term Examination, Odd Semester 2023-24

B. Tech. (Hons.) CS, II Year, III Semester

Subject Code: BCSC 0054

Subject Name: ADVANCED DATABASE MANAGEMENT SYSTEM

Time: 1 Hour **Maximum Marks: 15**

Instruction for students:

1. Mention clearly if any assumptions are being made.
2. No 'B' sheet will be given.

Section – A

*Attempt **All** Questions* 2 X 3 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Assume that you are appointed in your dream company as a database designer and got a project to work on backend implementation. Your job profile includes designing a complete database for a web based project. Identify and briefly explain all the phases you need to handle during the database development life cycle during this project.	2	1	U	C
2	*"Database Management System is preferred over File Systems".* Justify your answer with proper reasons and scenario.	2	1	U	C
3	Consider a relation having six attributes. Identify the maximum number of Super keys that can exist in this relation. Also Define Super Key.	2	3	R	P

Section – B

Attempt All Questions 3 X 3 = 9 Marks

No.	Detail of Question	Marks	CO	BL	KL
4	Consider the following schema: SUPPLIER (SID, SNAME, ADDRESS) PARTS (PID, PNAME, COLOR) CATALOG (SID, PID, COST) Write the Relational Algebra expression for the following queries: A. Find the name of all the suppliers who supply green parts. B. Find the average cost. C. Delete record of that supplier(s) whose name is 'Jatin'.	3	3	A	P
5	Consider the Relation R(WXYZ) and Functional Dependencies given below: FD: {WX → YZ, WY → XZ, XY → Z} Identify Candidate Key(s) and Prime attributes in this relation.	3	4	R	P
6	Consider the following Supplier-Parts-Projects database: SUPPLIER (SID, SNAME, STATUS, CITY) PARTS (PID, PNAME, COLOR, WEIGHT, CITY) PROJECT (PRID, PRNAME, CITY) SUPPLIED_QUANTITY (SID, PID, PRID, QTY) Draw its ER diagram.	3	3	An	P

Course Name: English for Professional Purposes I

Course Outcome:

The students will be able to:

CO1: create structurally effective, grammatically accurate and contextually appropriate written and spoken pieces for communication both independently and collaboratively.

CO2: display competence in all aspects of language learning not limited to listening, speaking, reading and writing.

CO3: develop independent perspectives and points of view and express them in a cohesive and meaningful way.

CO4: produce a repertoire of register-specific and academic vocabulary appropriate in both oral and written communication.

CO5: use language as an effective tool of communication for various contexts in both academic and professional setting.

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Mid Term Examination, Odd Semester 2023-24
B. Tech (All Branches), Year II, Semester III
BELH 1003 English for Professional Purposes I

Time: 2 Hours **Maximum Marks: 30**

Instruction for students: Follow the prescribed word limits.
Mention the correct option against properly numbered questions.
Do not copy the whole answer from the question paper.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Details of Questions	Marks	CO	BL	KL
1	Answer the following questions as directed. i- *My parents have decided to move to Delhi by the end of this year.* The non-finite verb in the given sentence is: a. have b. decided c. to move d. to Delhi ii- *I <u>love</u> you.* and *My <u>love</u> for you is eternal.* What part of speech does 'love' belong to in both these sentences? a. Verb b. Noun c. Verb and Noun d. Verb and Adjective iii- *The protesters were throwing <u>stones</u> at the police.* *The sculpture is made of <u>stone</u>.* What type of noun is 'stone' in the above sentences? a. Countable b. Uncountable c. Uncountable and Countable	3	1	A	F

	d. Countable and Uncountable iv- What would be the abstract noun form of the verb '*describe*'? a. described b. description c. decrepit d. discussion v- '*Alas!*', the interjection expresses: a. joy b. sadness c. surprise d. pain vi- *Soham and I stood by the window for some time and <u>neither</u> said anything.* What part of speech does 'neither' belong to? a. Conjunction b. Adjective c. Noun d. Pronoun vii- *One fine morning, the busy doctor invited Rohan into the hallway.* Identify the subject. a. morning b. the busy c. Rohan d. the busy doctor viii- Identify the form of verb underlined in the following sentence: *We <u>were</u> looking for a place to eat.* a. Transitive b. Intransitive c. Auxiliary d. Linking ix- Insert the appropriate preposition in the following sentence: *Let's have a break _____ noon.* a. at b. on c. in d. about				

	x- Identify the adjective in the following sentence: *The FA Cup has been won by the favourites whose favourite colour is red.* a. red b. favourites c. favourite d. Both b and c xi- I visit the Taj Mahal _____ I go to Agra. a. once b. whenever c. wherever d. and xii- Choose the grammatically correct sentence: a. My brother is good in English. b. My brother is good at English. c. both a & b are correct d. both a & b are incorrect				
2	i- Which of the following words cannot function as a determiner? a. some b. the c. when d. ten ii-Complete the following sentence with the correct determiner I am afraid we need to go shopping. There's very _____ milk left. a. little b. much c. a lot d. Some iii-Too _____ time is lost while haggling with _____ roadside vendor. a. more, more b. much, any c. much, the d. Both b and c are correct iv-Which of the following is a possessive determiner? a. it's b. Silpa's c. who's d. they're	3	1	A	F

v-In which of the following sentences is the word 'This' acting as a determiner?

A: *This* is my pen.
B: *This* pen is mine.

a. only A
b. only B
c. both A and B
d. neither A nor B

vi-Find the part with the error in the following sentence:

I will tell mine own story of success.

a. I will tell
b. mine own story
c. of success
d. No error

vii-Have you seen _____ orange pen that I like?

a. a
b. the
c. an
d. mine

viii-Identify the determiner(s) in the following sentence.

All the five committee members were present for the meeting.

a. all
b. five
c. the
d. a, b and c

ix-Can you please assist me for_____ days?

a. few
b. a few
c. little
d. many

In each of the questions given below, there are two segments labelled as (P) and (Q). Mark your answer as per the letter codes provided below:

A. Both P and Q are correct.
B. Both P and Q are incorrect.
C. P is correct but Q is incorrect.
D. P is incorrect but Q is correct.

x-(P): plenty of furniture (Q): plenty of student
xi-(P): twice the expense (Q): double the expense
xii- (P): much of his life (Q): most of his life

3	**Choose the most appropriate option/label for the portion in bold:** i. There's someone walking **up and down the drive**. a. Noun phrase b. Prepositional phrase c. Adjective phrase d. Adverb phrase e. None of these ii. Doubtless there was **a portrait of Lawrence of Arabia at the National Portrait Gallery**. a. Noun phrase b. Prepositional phrase c. Adjective phrase d. Adverb phrase e. None of these iii. We kept the new money **quite separately from what we'd already collected**. a. Verb phrase b. Prepositional phrase c. Adjective phrase d. Adverb phrase e. None of these iv. One might as well ask if the house itself **would have been** there. a. Noun phrase b. Prepositional phrase c. Verb phrase d. Adverb phrase e. None of these v. He thinks that he is **the best person to ever walk the earth**. a. Noun phrase b. Verb phrase c. Adjective phrase d. Adverb phrase e. None of these vi. **She might sing**, but I don't think she will. a. Subordinate clause b. Principal clause c. Noun phrase	3	1	A	F

(5)

d. Adjective phrase
e. None of the above

vii. Mary is dusting the furniture **because Alice won't**.
a. Subordinate clause
b. Principal clause
c. Adverb phrase
d. Adjective phrase
e. None of the above

viii. He began to laugh **quite loudly**.
a. Noun phrase
b. Prepositional phrase
c. Adjective phrase
d. Adverb phrase
e. None of these

ix. Dilip **might have been waiting** outside for you.
a. Principal clause
b. Subordinate clause
c. Verb phrase
d. Adverb phrase
e. None of these

x. **Poor though he is**, he's happy.
a. Adjective phrase
b. Principal clause
c. Adverb phrase
d. Subordinate clause
e. None of the above

xi. In Rome, when I thought I saw the cat, I was **undeniably mad.**
a. Noun phrase
b. Verb phrase
c. Adjective phrase
d. Adverb phrase
e. None of these

xii. I happen to believe **the story about Giotto and the circle,** by the way.
a. Noun phrase
b. Verb phrase
c. Adjective phrase
d. Adverb phrase

	e. None of these				
4	Read the following article and form your arguments in favour and against the topic: Gender contributes to Social Inequality You have to write three arguments in favour and three against the topic. **Many Faces of Gender Inequality** Inequality between women and men can appear in many different forms - it has many faces. Gender disparity is, in fact, not one affliction but a multitude of problems. Sometimes the different asymmetries are quite unrelated to each other. Indeed, there may be no significant inequality in one sphere but a great deal of inequality in another. For example, Japan has no particular gender bias in nutrition or health care or school education, but men do seem to have considerable relative advantage in securing high leadership positions in administration or business. However, in other cases, gender inequality of one type tends to encourage and sustain gender inequality of other kinds. Consequential analysis can then be critically important even within the large corpus of gender relations in general, in order to examine and scrutinize how the different aspects of gender inequality relate to each other. For while gender inequality has many faces, these are not independent (like those in the austere image of Brahma in early Indian iconography). Rather, they speak to each other and sometimes strongly encourage one another. For example, when women lack decisional power within the family, which amounts to a deprivation of women's effective agency, this can also adversely affect their own well-being. The two kinds of deprivation may, not only move together - be 'covariant' - but they may be linked with each other through causal connections.	3	4	An	C
5	Write a paragraph on the topic "Knowledge vs. Intelligence". (100 words)	3	2	U	C

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No	Details of Questions	Marks	CO	BL	KL
6	Identify the most suitable thought group boundaries in each of the following sentences and underline the focus word. a. Tamilnadu DGP Shanker Jiwal has reportedly ordered an inquiry after CM MK Stalin's convoy was stuck in traffic on its way back from Mamallapuram due to AR Rehman's concert.	5	2	A	C

	b. We are like books most people only see our covers the minority read only the introduction many people believe the critics few will know our content. c. An Indian Army colonel commanding a Rashtriya Rifles unit along with a Major were martyred in an encounter with terrorists in Anantnag in Kashmir. d. Us president Joe Biden's son Hunter Biden was criminally Charged on Thursday after efforts to reach a plea deal failed leading to first ever prosecution of a sitting president's child. e. A natural language interface usually has its own schema which is similar to the database conceptual schema, as well as a dictionary of important words.				
7	**A. Read the following passage and answer the questions that follow:** The phrase 'what is it like?' stands for a fundamental thought process. How does one go about observing and reporting on things and events that occupy the segments of earth space? Of all the infinite varieties of phenomena on the face of the earth, how does one decide what phenomena to observe? There is no such thing as a complete description of the earth or any part of it, as every microscopic point on the earth's surface differs from every other such point. Experience shows that the things observed are already familiar because they are like phenomenon that occur at home or because they resemble the abstract images and models developed in the human mind. How are abstract images formed? Humans alone, amongst all other animals on the earth, possess language, and their words symbolize not only specific things, but also mental images of classes of things. People can remember what they have seen or experienced because they attach a word symbol to them. During the long record of our efforts to gain more and more knowledge about the face of the earth as the human habitat, there has been a continuing interplay between things and events. The direct observation through the senses is described as a percept, and the mental image is described as a concept. Percepts are what some people describe as reality, in contrast to mental images, which are theoretical, implying that they are not real. The relation of percept to concept is not as simple as the definition implies. It is now quite clear that people of different	2.5+2.5	2	An	F

	cultures or even individuals in the same culture develop different mental images of reality and what they perceive is a reflection of these preconceptions. The direct observation of things and events on the face of the earth is therefore, clearly a function of the mental images of the mind of the observer that the whole idea of reality must be reconsidered. Concepts determine what the observer perceives, yet concepts are derived from the generalizations of previous percepts. What happens is that the educated observer is taught to accept a set of concepts and then, he/she sharpens or changes these concepts during their professional career. In any field of scholarship, professional opinion at one time determines what concepts and procedures are acceptable, and these form a kind of model of scholarly behaviour. i. The problem raised in the passage reflects on: A. Thought process B. Human behaviour C. Cultural perceptions D. Professional opinion ii. According to the passage, human beings have which of the following in mind the most? A. Observation of things B. Preparation of mental images C. Expression through language D. To gain knowledge iii. 'Concept' means: A. Mental image B. Reality C. An idea expressed in language D. All of the above iv. The relation of percept to concept is: A. Positive B. Negative C. Reflective D. Absolute v. In the passage, the Earth is taken as A. The globe B. The human habitat				

	C. A celestial body D. A planet vi. How are experiences resembled in the human mind? A. Abstract images B. Abstract models C. Symbols of specific things D. Both A and B vii. _____ is described as a direct observation through senses which is what people define as reality. A. Concept B. Percept C. Mental image D. Both B and C viii. What is a model of scholarly behaviour? A. Procedures B. Professional opinion C. Mental images D. Direct observation ix. A function of mental images of the mind of the observer about the reality is _____. A. Long record of our efforts B. Direct observation of things C. Direct observation of events D. Both B and C x. Which among the following does not mean Scholarly A. Academic B. Pedagogic C. Benighted D. Studious					
8	**A. Read the following paragraph and** 1-Identify the method: inductive or deductive and justify your answer. 2-Write down the transition words used in the paragraph. 3-Is the paragraph coherent? Explain how? *The road is one of the great fundamental institutions of mankind. Not only is the road one of the great human institutions because it is fundamental to social existence, but*	3+2	1,4	A,R	F	

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also because its varied effects appear in every department of the State. It is the road which determines the sites of many cities and the growth and nourishment of all. It is the road which controls the development of strategies and fixes the sites of battles. It is the road the gives its framework to all economic development. It is the road which is the channel of all trade, and, what is more important, of all ideas. In its most humble function, it is a necessary guide without which progress from place to place would be a ceaseless experiment, it is a sustenance without which organised society would be impossible, the road moves and controls all history.

B. Fill in the blanks with most suitable option

i. _____ digging here in the 19th century found a burial cist in the centre, although there are no records of any other remains.

a. Antiquarians
b. Archetype
c. Reticent
d. contemporary

ii. "Because natural science can offer _____ proof for its hypotheses, it can verify its claims."

a. empirical
b. speculative
c. maiden
d. freelance

iii. All the members of my book club share a _____taste in reading so we never have any issues agreeing on a book.

a. homogenous
b. amateur
c. inexplicable
d. infallible

iv. They are demanding the _____ of ancient treasures that were removed from the country in the 16th century.

a. recall
b. restitution
c. reissue
d. simulation

v. The construction of a _____ spreadsheet will allow you to develop a scenario for the students to attempt.

a. stereotype
b. prototype

	c. simulation d. liquidation vi. We were never so _____ about these styles in our time; we did it without so much talking about it. a. versatile b. flamboyant c. hypocrite d. obsolete vii. If he is a good neighbour, he will apologise and not make such a _____ for the stolen things. a. decommission b. simulation c. restitution d. hullabaloo viii. Finally, through a _____ of corridors Rohan found Ruchi's office. a. servile b. labyrinthine c. fixture d. labyrinth				

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive). However, For Engg. Courses in addition to F, C, P & M include D-Fundamental Design Principles, S-Criteria and Specifications, PC-Practical Constraints, DI- Design Instrumentalities

Course Name: Computer Organization

Course Outcome

CO1- Understand the basics of digital computer system.
CO2- Demonstrate the principle of arithmetic operations on unsigned, signed integers and floating point numbers.
CO3- Understand the concepts of combinational and sequential circuits and their applications
CO4- Understand the CPU architecture and organization
CO5- Explain the basic concepts of pipelining
CO6- Design the steps for the execution of the complete instruction for hardwired and micro-programmed control unit.
CO7- Explain the function of memory hierarchy
CO8- Determine the interface of CPU with input/output devices and their modes of transfer.

Printed Pages: 02 **University Roll No. .....**

Mid Term Examination, Odd Semester 2023-24
B.Tech (CS/AIML/CCV/CSF/DA/IIOT), II, III Semester
Computer Organization (BCSC 1005)

Time: 2 Hours **Maximum Marks: 30**

Section – A

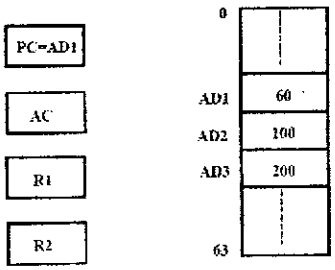
Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	In an arithmetic logic unit, two half subtractors are connected to form a full subtractor. Show the connections between half subtractors. Also, determine the required Boolean logic for the outputs.	3	3	U	C
2	Considering a 2's complement-based 8-bit processor, perform the following arithmetic operations. Also, check the existence of overflow in each operation. (a) (-100)+(15) (b) (-32)+(-97)	3	2	A	P
3	(a)A bus organization has eight 4-bit registers. Show, how multiplexers can be used to select registers in a common bus configuration? (b) Considering the above configuration, write the RTL statement of the micro-operation which will be executed upon the control unit's generation of the control word 01101000100101.	3	4	U	C
4	The initial content of 8-bit registers A, B, C, and D is given by A=11110011; B=10100111; C=11001101; D=11100010 What will be the content in each register after executing the following sequence of microoperations? A←$\bar{A} \vee \bar{B}$ D←A+$\bar{C}$+1	3	4	A	P

	C←ashr D				
5	How does a computer programme in 32-bit and 64-bit processors calculate the precise value of the distance between a spacecraft and a planet, if the distance is approximated by 101.765 meters?	3	2	A	P

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL
6	A digital signal processor uses the Booth's algorithm to implement a binary multiplier. Following the systematic multiplication process, show how the processor operates $(12) \times (-35)$.	5	4	A	P
7	(a) Using an accumulator-type computer with one instruction, write a programme to evaluate the following arithmetic statement. $$X = \frac{A - (B + C)}{D * E - F}$$ (b) Considering the following organization of the 64B memory and some processor registers, write the addressing mode of each of the following instructions.  (i) R1←M[AD1]+M[AD2] (ii) AC←M[PC+AD3] (iii) M[AD3]←R2 (iv) AC←M[M[AD1]] (v) M[AD1]←AC	5	4	U	C
8	A small portion of a 32MB memory is assigned for a stack and a processor register is used as a stack pointer. Following the postfix notation, how the following arithmetic expression will be evaluated by the stack? (((110/(30-8))*50)+5)-((15/(10-5))*2)	5	4	A	P

Course Name: English for Professional Purposes I

Course Outcome:

The students will be able to:

CO1: create structurally effective, grammatically accurate and contextually appropriate written and spoken pieces for communication both independently and collaboratively.

CO2: display competence in all aspects of language learning not limited to listening, speaking, reading and writing.

CO3: develop independent perspectives and points of view and express them in a cohesive and meaningful way.

CO4: produce a repertoire of register-specific and academic vocabulary appropriate in both oral and written communication.

CO5: use language as an effective tool of communication for various contexts in both academic and professional setting.

Printed Pages: 102 **University Roll No. .....**

Mid Term Examination, Odd Semester 2023-24
B. Tech (All Branches), Year II, Semester III
BELH 1003 English for Professional Purposes I

Time: 2 Hours **Maximum Marks: 30**

Instruction for students: Follow the prescribed word limits.
Mention the correct option against properly numbered questions.
Do not copy the whole answer from the question paper.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Details of Questions	Marks	CO	BL	KL
1	Answer the following questions as directed. i- *My parents have decided to move to Delhi by the end of this year.* The non-finite verb in the given sentence is: a. have b. decided c. to move d. to Delhi ii- *I <u>love</u> you.* and *My <u>love</u> for you is eternal.* What part of speech does 'love' belong to in both these sentences? a. Verb b. Noun c. Verb and Noun d. Verb and Adjective iii- *The protesters were throwing <u>stones</u> at the police.* *The sculpture is made of <u>stone</u>.* What type of noun is 'stone' in the above sentences? a. Countable b. Uncountable c. Uncountable and Countable	3	1	A	F

	d. Countable and Uncountable iv- What would be the abstract noun form of the verb '*describe*'? a. described b. description c. decrepit d. discussion v- '*Alas!*', the interjection expresses: a. joy b. sadness c. surprise d. pain vi- *Soham and I stood by the window for some time and <u>neither</u> said anything.* What part of speech does 'neither' belong to? a. Conjunction b. Adjective c. Noun d. Pronoun vii- *One fine morning, the busy doctor invited Rohan into the hallway.* Identify the subject. a. morning b. the busy c. Rohan d. the busy doctor viii- Identify the form of verb underlined in the following sentence: *We <u>were</u> looking for a place to eat.* a. Transitive b. Intransitive c. Auxiliary d. Linking ix- Insert the appropriate preposition in the following sentence: *Let's have a break _____ noon.* a. at b. on c. in d. about				

	x- Identify the adjective in the following sentence: *The FA Cup has been won by the favourites whose favourite colour is red.* a. red b. favourites c. favourite d. Both b and c xi- I visit the Taj Mahal _____ I go to Agra. a. once b. whenever c. wherever d. and xii- Choose the grammatically correct sentence: a. My brother is good in English. b. My brother is good at English. c. both a & b are correct d. both a & b are incorrect				
2	i- Which of the following words cannot function as a determiner? a. some b. the c. when d. ten ii-Complete the following sentence with the correct determiner I am afraid we need to go shopping. There's very _____ milk left. a. little b. much c. a lot d. Some iii-Too _____ time is lost while haggling with _____ roadside vendor. a. more, more b. much, any c. much, the d. Both b and c are correct iv-Which of the following is a possessive determiner? a. it's b. Silpa's c. who's d. they're	3	1	A	F

v-In which of the following sentences is the word 'This' acting as a determiner?

A: *This* is my pen.
B: *This* pen is mine.

a. only A
b. only B
c. both A and B
d. neither A nor B

vi-Find the part with the error in the following sentence:

I will tell mine own story of success.

a. I will tell
b. mine own story
c. of success
d. No error

vii-Have you seen _____ orange pen that I like?

a. a
b. the
c. an
d. mine

viii-Identify the determiner(s) in the following sentence.

All the five committee members were present for the meeting.

a. all
b. five
c. the
d. a, b and c

ix-Can you please assist me for_____ days?

a. few
b. a few
c. little
d. many

In each of the questions given below, there are two segments labelled as (P) and (Q). Mark your answer as per the letter codes provided below:

A. Both P and Q are correct.
B. Both P and Q are incorrect.
C. P is correct but Q is incorrect.
D. P is incorrect but Q is correct.

x-(P): plenty of furniture (Q): plenty of student
xi-(P): twice the expense (Q): double the expense
xii- (P): much of his life (Q): most of his life

3	**Choose the most appropriate option/label for the portion in bold:** i. There's someone walking **up and down the drive**. a. Noun phrase b. Prepositional phrase c. Adjective phrase d. Adverb phrase e. None of these ii. Doubtless there was **a portrait of Lawrence of Arabia at the National Portrait Gallery**. a. Noun phrase b. Prepositional phrase c. Adjective phrase d. Adverb phrase e. None of these iii. We kept the new money **quite separately from what we'd already collected**. a. Verb phrase b. Prepositional phrase c. Adjective phrase d. Adverb phrase e. None of these iv. One might as well ask if the house itself **would have been** there. a. Noun phrase b. Prepositional phrase c. Verb phrase d. Adverb phrase e. None of these v. He thinks that he is **the best person to ever walk the earth**. a. Noun phrase b. Verb phrase c. Adjective phrase d. Adverb phrase e. None of these vi. **She might sing**, but I don't think she will. a. Subordinate clause b. Principal clause c. Noun phrase	3	1	A	F

(5)

d. Adjective phrase
e. None of the above

vii. Mary is dusting the furniture **because Alice won't**.
a. Subordinate clause
b. Principal clause
c. Adverb phrase
d. Adjective phrase
e. None of the above

viii. He began to laugh **quite loudly**.
a. Noun phrase
b. Prepositional phrase
c. Adjective phrase
d. Adverb phrase
e. None of these

ix. Dilip **might have been waiting** outside for you.
a. Principal clause
b. Subordinate clause
c. Verb phrase
d. Adverb phrase
e. None of these

x. **Poor though he is**, he's happy.
a. Adjective phrase
b. Principal clause
c. Adverb phrase
d. Subordinate clause
e. None of the above

xi. In Rome, when I thought I saw the cat, I was **undeniably mad.**
a. Noun phrase
b. Verb phrase
c. Adjective phrase
d. Adverb phrase
e. None of these

xii. I happen to believe **the story about Giotto and the circle,** by the way.
a. Noun phrase
b. Verb phrase
c. Adjective phrase
d. Adverb phrase

	e. None of these				
4	Read the following article and form your arguments in favour and against the topic: Gender contributes to Social Inequality You have to write three arguments in favour and three against the topic. **Many Faces of Gender Inequality** Inequality between women and men can appear in many different forms - it has many faces. Gender disparity is, in fact, not one affliction but a multitude of problems. Sometimes the different asymmetries are quite unrelated to each other. Indeed, there may be no significant inequality in one sphere but a great deal of inequality in another. For example, Japan has no particular gender bias in nutrition or health care or school education, but men do seem to have considerable relative advantage in securing high leadership positions in administration or business. However, in other cases, gender inequality of one type tends to encourage and sustain gender inequality of other kinds. Consequential analysis can then be critically important even within the large corpus of gender relations in general, in order to examine and scrutinize how the different aspects of gender inequality relate to each other. For while gender inequality has many faces, these are not independent (like those in the austere image of Brahma in early Indian iconography). Rather, they speak to each other and sometimes strongly encourage one another. For example, when women lack decisional power within the family, which amounts to a deprivation of women's effective agency, this can also adversely affect their own well-being. The two kinds of deprivation may, not only move together - be 'covariant' - but they may be linked with each other through causal connections.	3	4	An	C
5	Write a paragraph on the topic "Knowledge vs. Intelligence". (100 words)	3	2	U	C

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No	Details of Questions	Marks	CO	BL	KL
6	Identify the most suitable thought group boundaries in each of the following sentences and underline the focus word. a. Tamilnadu DGP Shanker Jiwal has reportedly ordered an inquiry after CM MK Stalin's convoy was stuck in traffic on its way back from Mamallapuram due to AR Rehman's concert.	5	2	A	C

	b. We are like books most people only see our covers the minority read only the introduction many people believe the critics few will know our content. c. An Indian Army colonel commanding a Rashtriya Rifles unit along with a Major were martyred in an encounter with terrorists in Anantnag in Kashmir. d. Us president Joe Biden's son Hunter Biden was criminally Charged on Thursday after efforts to reach a plea deal failed leading to first ever prosecution of a sitting president's child. e. A natural language interface usually has its own schema which is similar to the database conceptual schema, as well as a dictionary of important words.				
7	**A. Read the following passage and answer the questions that follow:** The phrase 'what is it like?' stands for a fundamental thought process. How does one go about observing and reporting on things and events that occupy the segments of earth space? Of all the infinite varieties of phenomena on the face of the earth, how does one decide what phenomena to observe? There is no such thing as a complete description of the earth or any part of it, as every microscopic point on the earth's surface differs from every other such point. Experience shows that the things observed are already familiar because they are like phenomenon that occur at home or because they resemble the abstract images and models developed in the human mind. How are abstract images formed? Humans alone, amongst all other animals on the earth, possess language, and their words symbolize not only specific things, but also mental images of classes of things. People can remember what they have seen or experienced because they attach a word symbol to them. During the long record of our efforts to gain more and more knowledge about the face of the earth as the human habitat, there has been a continuing interplay between things and events. The direct observation through the senses is described as a percept, and the mental image is described as a concept. Percepts are what some people describe as reality, in contrast to mental images, which are theoretical, implying that they are not real. The relation of percept to concept is not as simple as the definition implies. It is now quite clear that people of different	2.5+2.5	2	An	F

	cultures or even individuals in the same culture develop different mental images of reality and what they perceive is a reflection of these preconceptions. The direct observation of things and events on the face of the earth is therefore, clearly a function of the mental images of the mind of the observer that the whole idea of reality must be reconsidered. Concepts determine what the observer perceives, yet concepts are derived from the generalizations of previous percepts. What happens is that the educated observer is taught to accept a set of concepts and then, he/she sharpens or changes these concepts during their professional career. In any field of scholarship, professional opinion at one time determines what concepts and procedures are acceptable, and these form a kind of model of scholarly behaviour. i. The problem raised in the passage reflects on: A. Thought process B. Human behaviour C. Cultural perceptions D. Professional opinion ii. According to the passage, human beings have which of the following in mind the most? A. Observation of things B. Preparation of mental images C. Expression through language D. To gain knowledge iii. 'Concept' means: A. Mental image B. Reality C. An idea expressed in language D. All of the above iv. The relation of percept to concept is: A. Positive B. Negative C. Reflective D. Absolute v. In the passage, the Earth is taken as A. The globe B. The human habitat				

	C. A celestial body D. A planet vi. How are experiences resembled in the human mind? A. Abstract images B. Abstract models C. Symbols of specific things D. Both A and B vii. _____ is described as a direct observation through senses which is what people define as reality. A. Concept B. Percept C. Mental image D. Both B and C viii. What is a model of scholarly behaviour? A. Procedures B. Professional opinion C. Mental images D. Direct observation ix. A function of mental images of the mind of the observer about the reality is _____. A. Long record of our efforts B. Direct observation of things C. Direct observation of events D. Both B and C x. Which among the following does not mean Scholarly A. Academic B. Pedagogic C. Benighted D. Studious					
8	**A. Read the following paragraph and** 1-Identify the method: inductive or deductive and justify your answer. 2-Write down the transition words used in the paragraph. 3-Is the paragraph coherent? Explain how? *The road is one of the great fundamental institutions of mankind. Not only is the road one of the great human institutions because it is fundamental to social existence, but*	3+2	1,4	A,R	F	

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also because its varied effects appear in every department of the State. It is the road which determines the sites of many cities and the growth and nourishment of all. It is the road which controls the development of strategies and fixes the sites of battles. It is the road the gives its framework to all economic development. It is the road which is the channel of all trade, and, what is more important, of all ideas. In its most humble function, it is a necessary guide without which progress from place to place would be a ceaseless experiment, it is a sustenance without which organised society would be impossible, the road moves and controls all history.

B. Fill in the blanks with most suitable option

i. _____ digging here in the 19th century found a burial cist in the centre, although there are no records of any other remains.

a. Antiquarians
b. Archetype
c. Reticent
d. contemporary

ii. "Because natural science can offer _____ proof for its hypotheses, it can verify its claims."

a. empirical
b. speculative
c. maiden
d. freelance

iii. All the members of my book club share a _____taste in reading so we never have any issues agreeing on a book.

a. homogenous
b. amateur
c. inexplicable
d. infallible

iv. They are demanding the _____ of ancient treasures that were removed from the country in the 16th century.

a. recall
b. restitution
c. reissue
d. simulation

v. The construction of a _____ spreadsheet will allow you to develop a scenario for the students to attempt.

a. stereotype
b. prototype

	c. simulation d. liquidation vi. We were never so _____ about these styles in our time; we did it without so much talking about it. a. versatile b. flamboyant c. hypocrite d. obsolete vii. If he is a good neighbour, he will apologise and not make such a _____ for the stolen things. a. decommission b. simulation c. restitution d. hullabaloo viii. Finally, through a _____ of corridors Rohan found Ruchi's office. a. servile b. labyrinthine c. fixture d. labyrinth				

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive). However, For Engg. Courses in addition to F, C, P & M include D-Fundamental Design Principles, S-Criteria and Specifications, PC-Practical Constraints, DI- Design Instrumentalities

Printed Pages: 16 University Roll No.....

Mid-Term Examination, Odd Semester 2023-24

B.Tech. CSE, II: Year, III: Semester

BCSE 0902: Smart Industrial Connectivity

Time: 1-Hour Maximum Marks: 30

Note: Attempt all questions. 60 x 0.5 = 30 Marks

Read the following code and give the answer mentioned below from 1 to 5

```
#include <SPI.h>
#include <mcp2515.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27,16,2
struct can_frame canMsg;
MCP2515 mcp2515(10);

void setup()
{
  Serial.begin(9600);
  SPI.begin();
  lcd.init();
  lcd.clear();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("CANBUS TUTORIAL");
  delay(3000);
  lcd.clear();
```

```
  mcp2515.reset();
  mcp2515.setBitrate(CAN_500KBPS, MCP_8MHZ);
  mcp2515.setNormalMode();
 }
 void loop()
 {
  if (mcp2515.readMessage(&canMsg) == MCP2515::ERROR_OK) //
To receive data (Poll Read)
  {
   int x = canMsg.data[0];
   int y = canMsg.data[1];

   lcd.setCursor(0, 0);
   lcd.print("Humi: ");
   lcd.print(x);
   lcd.setCursor(0, 1);
   lcd.print("Temp: ");
   lcd.print(y);
   delay(1000);
   lcd.clear();
  }
}
```

1. What is the purpose of the command "mcp2515.setNormalMode()" in the setup function?

A. To set the CAN communication in normal mode
B. To set the controller area network in sleep mode
C. To set the bitrate of CAN communication
D. To begin SPI communication

2. What is the purpose of the function "mcp2515.readMessage(&canMsg)"?

A. To change the bitrate of CAN communication
B. To reset the MCP2515 controller
C. C.To read the incoming data from CAN communication
D. To send data through the CAN communication

3. What are the arguments passed to the function "mcp2515.setBitrate(CAN_500KBPS, MCP_8MHZ)"?

A. The speed and clock of the CAN communication
B. The MCP2515 controller and SPI communication
C. C.The speed and clock of the MCP2515 controller
D. The MCP2515 controller and CAN communication

4. What is the purpose of including the libraries "SPI.h" and "mcp2515.h"?

A. To use CAN communication
B. To use Serial Communication
C. To use I2C communication
D. To use Ethernet Communication

5. What is the purpose of the command "lcd.backlight()"?

A. None of the above
B. To turn off the backlight of the LCD display
C. To turn on the backlight of the LCD display
D. To increase the temperature and humidity reading accuracy

6. Which communication protocol is commonly used for real-time automation and control within industrial systems?

A. HTTP

B. FTP
C. Modbus
D. SMTP

7. Which of the following best describes the role of an Industrial Gateway template in ThingWorx?
   A. It enables communication between ThingWorx and industrial devices using a specific protocol.
   B. It controls the lighting and HVAC systems in industrial facilities.
   C. It manages user authentication and access control in ThingWorx.
   D. It creates virtual reality simulations for industrial processes.

8. What is typically included in an Industrial Gateway template configuration?
   A. Employee schedules and work assignments
   B. Device-specific settings, such as communication protocols and data mapping
   C. Marketing materials for industrial products
   D. Recipes for cooking industrial-grade meals

9. In ThingWorx, how does an Industrial Gateway template help streamline the integration of industrial devices?
   A. By providing on-site technical support to device operators
   B. By automatically diagnosing and repairing device issues
   C. By offering financial incentives to device manufacturers
   D. By offering pre-defined configurations and templates for various industrial devices

10. What is the benefit of using Industrial Gateway templates in ThingWorx for IoT projects?

A. It reduces the need for industrial devices altogether.
B. It simplifies and accelerates the integration of diverse industrial devices.
C. It replaces the need for industrial automation systems.
D. It automates employee training processes.

11. What is Zigbee?

A. A type of wireless security protocol
B. A high-speed internet connection
C. A low-power, wireless communication protocol
D. A brand of Bluetooth headphones

12. Which frequency bands are commonly used for Zigbee communication?

A. 2.4 GHz and 5 GHz
B. 900 MHz and 2.4 GHz
C. 5 GHz and 60 GHz
D. 433 MHz and 868 MHz

13. Zigbee is primarily designed for:

A. High-speed data transfer
B. Long-range communication
C. Low-power, short-range communication
D. Satellite communication

14. What is the maximum data rate typically supported by Zigbee?

A. 1 Mbps
B. 10 Mbps

C. 100 Kbps
D. 1 Kbps

15. Which of the following is NOT a common application for Zigbee technology?
A. Home automation
B. Industrial control
C. Wi-Fi-networking
D. Healthcare monitoring

16. Zigbee devices are known for their:
A. High power consumption
B. Large form factor
C. Scalability and interoperability
D. Expensive price

17. Which Zigbee topology is often used for building automation and sensor networks?
A. Star
B. Mesh
C. Point-to-point
D. Bus

18. What is the typical range of a Zigbee network in an indoor environment?
A. Up to 100 meters
B. Up to 1 kilometer
C. Up to 10 meters
D. Up to 5 kilometers

19. Which IEEE standard is associated with Zigbee?
    A. IEEE 802.11
    B. IEEE 802.15
    C. IEEE 802.3
    D. IEEE 802.16

20. Zigbee security is enhanced through the use of:
    A. Strong encryption and authentication
    B. Wide frequency bands
    C. Large antennas
    D. High data rates

21. What does CAN stand for in the context of communication protocols?
    A. Computer Area Network
    B. Controller Area Network
    C. Centralized Area Network
    D. Communication Area Network

22. Which communication standard does the HC-05 Bluetooth module use?
    A. NFC
    B. Zigbee
    C. Bluetooth
    D. Wi-Fi

23. Which of the following is true about CAN communication?
    A. It is primarily used for long-distance wireless

communication.

B. It is a point-to-point communication protocol.

C. It is commonly used in automotive and industrial applications.

D. It is a high-speed, optical communication protocol.

24. The MCP2515 CAN controller operates at which voltage level?

A. 3.3V

B. 5V

C. 12V

D. 24V

25. Which of the following is not a typical feature of the MCP2515 CAN controller?

A. SPI interface

B. Support for multiple message buffers

C. LIN communication support

D. Programmable bit rates

26. CAN use which type of communication bus?

A. Serial bus

B. Parallel bus

C. Optical bus

D. USB bus

27. In CAN communication, what is the purpose of the CAN identifier?

A. To specify the data length

B. To indicate the sender's address

C. To identify the type of message

D. To establish a secure connection

28. What is the maximum data rate supported by the MCP2515 CAN controller?

A. 100 Kbps
B. 1 Mbps
C. 10 Mbps
D. 100 Mbps

29. Which mode of operation allows the MCP2515 to transmit and receive messages simultaneously?

A. Normal mode
B. Loopback mode
C. Sleep mode
D. Listen-only mode

30. CAN communication is known for its:

A. Low data integrity
B. High data transfer latency
C. Robustness in noisy environments
D. Limited scalability

31. What is the HC-05?

A. A microcontroller
B. A Bluetooth module
C. A GPS module
D. A Wi-Fi module

32. What is the primary application of the MCP2515 CAN controller?

A. Audio processing

B. Ethernet communication
C. Automotive and industrial communication
D. Wireless networking

33. What is the HC-05's primary function?
A. Detecting temperature
B. Playing music
C. Wireless data communication
D. Controlling servos

34. What is the maximum range typically supported by the HC-05 module?
A. 10 meters
B. 50 meters
C. 100 meters
D. 1000 meters

35. Which communication modes are available with the HC-05 module?
A. Master and slave
B. Receiver and transmitter
C. Primary and secondary
D. Bluetooth 1 and Bluetooth 2

36. What is the default baud rate for the HC-05 module when using AT commands?
A. 9600 bps
B. 115200 bps
C. 57600 bps
D. 38400 bps

37. Which type of voltage levels does the HC-05 module typically operate with?

A. 5V
B. 3.3V
C. 12V
D. 24V

38. What is the role of the key pin (EN) on the HC-05 module?

A. It enables or disables the module
B. It selects the baud rate
C. It sets the Bluetooth device name
D. It adjusts the transmit power

39. The HC-05 module can be used for:

A. GPS navigation
B. Wireless charging
C. Home automation
D. FM radio broadcasting

40. Which AT command is typically used to change the Bluetooth device name on the HC-05 module?

A. AT+NAME
B. AT+VERSION
C. AT+BAUD
D. AT+ROLE

41. Which programming language is commonly used for custom scripting in Kepware software?

A. Python

B. JavaScript
C. C++
D. Ruby

42. Which company owns and develops Kepware software?
A. Microsoft
B. Siemens
C. Rockwell Automation
D. PTC (Parametric Technology Corporation)

43. What role does ThingWorx play in the context of IoT?
A. It's a communication protocol
B. It's an industrial robot
C. It's an IoT platform for building applications and services
D. It's a virtual reality simulator

44. How does Kepware software integrate with ThingWorx?
A. Through direct hardware connections
B. By using a middleware layer
C. It doesn't integrate with ThingWorx
D. By generating QR codes

45. What is the primary benefit of connecting Kepware software to ThingWorx?
A. Improved weather forecasting
B. Enhanced industrial automation and monitoring
C. Faster internet browsing
D. Better healthcare services

46. Which communication protocols are commonly supported by Kepware software for data acquisition?

A. HTTP and FTP
B. MQTT and CoAP
C. OPC and Modbus
D. SMTP and POP3

47. In the context of Kepware and ThingWorx integration, what does "IoT" stand for?

A. Internet of Things
B. Input/Output Transfer
C. Integrated Online Technology
D. Industrial Object Tracking

48. What is the typical use case for Kepware and ThingWorx integration?

A. Online gaming
B. Supply chain management
C. Remote patient monitoring
D. Industrial automation and predictive maintenance

49. What is Kepware software primarily used for in an industrial context?

A. Video surveillance
B. Industrial data connectivity and communication
C. Accounting and payroll
D. 3D modeling

50. What is the primary function of ThingWorx in the context of industrial connectivity?

A. It acts as a manufacturing machine
B. It serves as an industrial database

C. It provides IoT platform capabilities for data analysis and visualization
D. It controls traffic lights

51. Which wireless communication protocol is commonly used in industrial settings for short-range connectivity?
A. Wi-Fi
B. Bluetooth
C. Zigbee
D. NFC

52. What is the main purpose of SCADA (Supervisory Control and Data Acquisition) systems?
A. Social Communication and Data Analysis
B. Security and Control of Accessible Devices and Applications
C. Supervision and Remote Control of Industrial Processes
D. Systematic Communication and Data Aggregation

53. Which communication technology is suitable for long-range, low-power applications in industrial environments?
A. Wi-Fi
B. Cellular (4G/5G)
C. LoRaWAN
D. Bluetooth

54. What is OPC UA (Unified Architecture)?
A. Online Production Connectivity Unified Architecture
B. Open Platform Communications Unified Automation
C. Operational Protocol Communication Unified Architecture

D. Object-oriented Process Control Unified Approach

55. What is a common challenge in implementing industrial IoT solutions related to legacy equipment?

A. They are too expensive to integrate
B. They lack connectivity options
C. They are too modern for integration
D. They are not used in industrial settings

56. Which of the following is a key benefit of edge computing in industrial settings?

A. It reduces the need for cloud services
B. It increases latency for real-time applications
C. It decreases security concerns
D. It centralizes all data processing

57. Which network topology is characterized by a single central node connected to multiple subordinate nodes?

A. Mesh
B. Star
C. Bus
D. Ring

58. What does MQTT stand for in the context of industrial connectivity?

A. Message Queuing Telemetry Transport
B. Multiple Query Transmission Technique
C. Managed Quality Test Technology
D. Mobile Quick Transfer Tool

59. Which security concern is specific to industrial networks rather than traditional IT networks?

A. Malware attacks
B. Phishing emails
C. Data breaches
D. Physical sabotage

60. What is the primary purpose of an Industrial Gateway template in ThingWorx?

A. To design user interfaces for industrial applications
B. To store and manage historical data from industrial devices
C. To provide a standardized configuration for connecting industrial devices to ThingWorx
D. To monitor employee activity on the factory floor

Course Name: Statistics and Complex Analysis

Course Outcome

CO 1: Compute moments and use them to visualize symmetry and peakedness of the data
CO 2: Fit a straight line on the given data by the method of Least squares
CO 3: Measure the extent of correlation between two variables
CO 4: Predict the value of a variable by finding regression lines
CO 5: Apply discrete probability distributions to the real-life data problems
CO 6: Apply Chi-square test to test a statistical hypothesis in real-life data analysis
CO 7: Know the concepts of analytic and harmonic functions and their applications
CO 8: Solve problems involving complex integration
CO 9: Find radius and region of convergence of a power series
CO 10: Calculate residue and apply Cauchy's residue theorem in solving real integrals

Printed Pages: 3 **University Roll No. .....**

Mid Term Examination, Odd Semester 2023-24
B. Tech. (CS, ME), II Year, III Semester
Sub. Code & Sub. Name- BMAS 0106, Statistics and Complex Analysis

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:

1. Attempt ALL sections.
2. The symbols have their usual meanings.

Section – A

Attempt All Questions. 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	The first four moments about mean of a frequency distribution are **0, 60, −50** and **8020** respectively. Comment upon the Skewness and Kurtosis of the distribution.	3	1	R	C
2	By the method of Least Squares, find the straight line that best fits the following data: x: 1, 2, 3, 4, 5 y: 14, 27, 40, 55, 68	3	2	U	F

3	A coin is tossed six times. What is the probability of getting four or more heads?	3	5	A	P
4	Calculate the coefficient of correlation between marks obtained by 8 students in Mathematics and Statistics: **Student** A B C D E F G H **Mathematics** 25 30 32 35 37 40 42 45 **Statistics** 08 10 15 17 20 23 24 25	3	3	E	P
5	Suppose that a book of 600 pages contains 40 printing mistakes. Assume that these errors are randomly distributed throughout the book and x, the number of errors per page has a Poisson distribution. What is the probability that 10 pages selected at random will be free of errors?	3	5	An	C

Section – B

Attempt All Questions. 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	In a music competition, ten competitors were ranked by three Judges A, B and C in the following order: Judge A 5 3 10 7 2 1 4 10 4 6 Judge B 1 6 5 10 3 2 4 9 7 8 Judge C 6 4 9 8 1 2 3 10 5 7 Use the rank correlation coefficient to determine which pair of Judges has the nearest approach to common taste in music?	5	3	E	M
7	Two lines of regression are given by $x + 2y - 5 = 0$ and $2x + 3y - 8 = 0$. Also given that: $\sigma_x^2 = 12$. Calculate: (i) Mean of x and mean of y (ii) The coefficient of correlation between x and y. (iii) The standard deviation of y.	5	4	A	C

8	The following table gives the number of accounting clerks, committing errors and not committing errors, among trained and untrained clerks working in an organization. Test the effectiveness of training in preventing errors. Given that: $\chi^2(0.05) = 3.84$ for 1 degree of freedom. **OR,** A shop owner claims that an equal number of customers come into his shop each weekday (except Saturday and Sunday). To test this hypothesis, an independent researcher records the number of customers that come in to the shop on a given week and find the following observations: **Monday: 50 customers** **Tuesday: 60 customers** **Wednesday: 40 customers** **Thursday: 47 customers** **Friday: 53 customers** Perform a Chi-square good of fit test to determine if the data is consistent with the shop owner's claim? [Given that: The tabulated value of χ^2 at 5 % level of significance for 4 degree of freedom is 9.488].	5	6	C	M

	Number of clerks committing errors	Number of clerks not committing errors	Total
Trained	**70**	**530**	**600**
Untrained	**155**	**745**	**900**
Total	**225**	**1275**	**1500**

Course Name: Operating System

Course Outcome

CO1 - Understand the classification of operating system environment
CO2 - Understand the basic of process management
CO3 - Apply the concept of CPU process scheduling for the given scenarios.
CO4 - Illustrate the process synchronization and concurrency process in operating system.
CO5 - Analyze the occurrence of deadlock in operating system
CO6 - Describe and analyze the memory management and its allocation policies.
CO7 - Understand the concepts of disk scheduling

Printed Pages: 03 **University Roll No.** .....

Mid Term Examination, Odd Semester 2023-24

B. Tech- CSE/ EC with Minor CS/AIML/CCV/DA/CSF/IIOT, II Year III Semester

Operating System (BCSC 0004)

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	i) Give two reasons why caches are useful. What problems do they solve? ii) What is the main difficulty that a programmer must overcome in writing an operating system for a real-time environment?	1+2	1	An	C
2	Discuss the merits and demerits of following operating systems: a) Multiprogramming b) Multitasking	1.5+1.5	1	U	F
3	Consider three CPU-bound processes arriving to a First Come First Served (FCFS) scheduler as follows: **Process** **Arrival time** **Burst time** P1 0 2 P2 1 12 P3 5 1 i) Calculate the average waiting time across all three processes. ii) A weakness of FCFS is that its performance is sensitive to the arrival process. Assume process arrival times remain t = 0, 1, 5 but the arrival order changes. Give an alternative arrival order that improves the average waiting time. Define	1+2	3	A	P

	the convoy effect and give a second alternative arrival order displaying it.				
4	Describe the five-state process model, describe what transitions are valid between the five states, and describe an event that might cause such a transition.	3	2	U	P
5	You are a developer working on a multi-threaded application. The application frequently encounters race conditions, leading to unpredictable behavior and crashes. a) Define what race conditions are and why they occur in multi-threaded applications? b) Discuss synchronization mechanisms and best practices you would use to eliminate race conditions and ensure the application behaves predictably.	1+2	4	An	M

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	i) You are tasked with simulating the Dining Philosophers problem, where five philosophers sit around a circular table, and they alternate between thinking and eating. To eat, a philosopher must pick up two forks placed between them and their neighbors. Describe the challenges and potential solutions to this problem using Semaphore. ii) If we add a 6th chopstick to the center of the table, have we cured the deadlock problem? If yes, what condition have we removed? If no, explain why not.	4+1	4	A	P
7	A shared variable x, initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to memory, and then terminates. Each of the processes Y and Z	5	4	An	P

	reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?				
8	Consider a uniprocessor system executing three tasks T1, T2 and T3, each of which is composed of an infinite sequence of jobs (or instances) which arrive periodically at intervals of 3, 7 and 20 milliseconds, respectively. The priority of each task is the inverse of its period and the available tasks are scheduled in order of priority, with the highest priority task scheduled first. Each instance of T1, T2 and T3 requires an execution time of 1, 2 and 4 milliseconds, respectively. Given that all tasks initially arrive at the beginning of the 1st milliseconds and task preemptions are allowed. In how many milliseconds the first instance of T3 completes its execution?	5	3	E	M

Course Name: Introduction to Machine Learning

Course Outcome

- CO1: Apply the basic concepts of machine learning including bias-variance tradeoff.
- CO2: Apply the concepts of regression.
- CO3: Conceptualize supervise and re-enforcement learning for classification.
- CO4: Formulate the ensemble methods for improving classification.
- CO5: Apply ANN with optimization in machine learning.
- CO6: Design and develop projects based on machine learning.

Printed Pages: **University Roll No. .....**

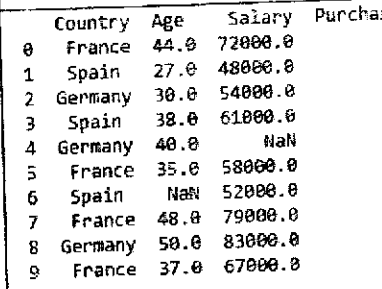
Mid Term Examination, Odd Semester 2023-24
B.Tech(AIML), 2nd Year, 3rd Semester
Subject Code -BCSC 0701
Subject Name-Introduction to Machine Learning

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	 Assume you are the supermarket manager of NIKE shoes and the above is the last month data of a top-	3	3	An	P

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	0
1	Spain	27.0	48000.0	1
2	Germany	30.0	54000.0	0
3	Spain	38.0	61000.0	0
4	Germany	40.0	NaN	1
5	France	35.0	58000.0	1
6	Spain	NaN	52000.0	0
7	France	48.0	79000.0	1
8	Germany	50.0	83000.0	0
9	France	37.0	67000.0	1

	notch quality sneaker purchased by players of different origin and of different age group. This data is not fit for direct ML analysis. Describes the steps that you will be taking to make it ready for analysis.				
2	Suppose you have a real-world dataset related to sales forecasting. Discuss the steps involved in splitting the dataset into training and test sets for machine learning purposes. Explain the importance of this data splitting process.	3	1	R	F
3	You work as a data analyst for an online retailer, "TechMart," which sells a variety of electronic gadgets. The company wants to optimize its inventory management by analyzing daily sales data for the past week. To do this, you plan to use NumPy to calculate the average daily sales. In the context of optimizing inventory management for TechMart, you have a NumPy array representing daily sales of electronic gadgets for the past week. Each element in the array corresponds to the number of units sold on a specific day. daily_sales = [50, 60, 75, 40, 55, 65, 70] Calculate and explain how the average daily sales, stored in 'average_sales', can help TechMart in inventory management. Write numpy code for the above program.	3	1	A	P
4	Explain how k-fold cross-validation can help mitigate the problems of overfitting and underfitting in machine learning models. Provide a brief description of the process, mentioning the impact on model assessment and selection.	3	1	A	P
5	In a school's student grading system, the administration wants to evaluate the performance of a newly introduced grading policy using a confusion matrix. The policy aims to categorize students'	3	3	U	C

	academic progress as either "Satisfactory" or "Needs Improvement" based on their test scores. Question: In the context of the school's grading system, explain what a confusion matrix is and why it is used. Provide a brief description of the key components of a confusion matrix.				

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Scenario: Predictive Maintenance in Manufacturing You are a data scientist working for a manufacturing company that produces heavy machinery. The company is interested in implementing predictive maintenance to reduce downtime and maintenance costs. Your task is to develop a machine learning model for predicting when a particular machine will likely require maintenance based on historical data. Question: In the context of predictive maintenance for heavy machinery, outline the key steps involved in building a machine learning model.	5	1	R	F
7	Imagine you are working as a data scientist for a real estate company that wants to predict house prices based on various factors. Consider the following scenario: a) The company is interested in understanding how the size of a house, measured in square feet (X1), impacts its selling price (Y). Explain how you would use simple linear regression to address this scenario. Describe the key components of the regression model and its interpretation. b) The company realizes that the number of bedrooms (X2) could also influence house prices. How would you extend the analysis to include the number of bedrooms as an additional predictor? Describe the multiple linear regression model in this	5	2	U	P

	context and provide the general form of the regression equation. c) During your analysis, you observe a high degree of correlation between the square footage (X1) and the number of bedrooms (X2). How might multicollinearity affect your multiple linear regression model's results? Explain the potential challenges and how you would address them. d) The company's stakeholders are interested in understanding how well your regression models fit the data. Describe how you would use the coefficient of determination (R-squared) to assess the goodness of fit for both the simple linear regression model (X1 predicting Y) and the multiple linear regression model (X1, X2 predicting Y). Interpret the meaning of R-squared values. e) As you develop increasingly complex regression models with more predictor variables, you start to encounter the risk of overfitting. Explain what overfitting is, how it can occur, and discuss strategies to prevent or mitigate overfitting when using linear regression for predictive modelling in this real estate context.				
8	Imagine you are a data scientist working for a medical diagnostics company developing a machine learning model to detect a rare medical condition. The dataset you are using is highly imbalanced, with only 2% of cases being positive for the condition. Evaluate the performance measures used in machine learning, focusing on accuracy and the confusion matrix, in the context of your medical diagnostic model. 1. Explain why using accuracy as the sole performance metric might not be appropriate for assessing the model's effectiveness in this scenario. 2. Discuss the potential consequences of relying on accuracy when dealing with imbalanced datasets and a rare medical condition.	5	1	E	C

	3. Provide an alternative performance metric or metrics that would be more suitable for evaluating the model's performance, considering the importance of correctly identifying positive cases.				

Course Name: Introduction to Virtualization and Cloud Computing

Course Outcome

CO1- Understand the basics of virtualization technology and hypervisors.
CO2- Analyze the trade-offs between deploying applications in the cloud and over the local infrastructure.
CO3- Recognize real-world problem using cloud computing through group collaboration
CO4- Understand on-demand utility computing phenomenon of cloud computing.
CO5- Understand on-demand utility computing phenomenon of cloud computing.

Printed Pages:02 **University Roll No. .....**

Mid Term Examination, Odd Semester 2023-24
B.Tech CCV, II Year, III Semester
BCSE 1502 Introduction to Virtualization and Cloud Computing

Time: 2 Hours **Maximum Marks: 30**

Instruction for students: Attempt all questions of a section at one place.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	A large enterprise is planning to upgrade its IT infrastructure. They want to understand how virtualization can be applied in different areas, such as servers, storage, and networks. Provide an overview of the various types of virtualizations applicable to each area and the corresponding technologies involved.	3	1	A	C
2	You are delivering a presentation to a group of IT professionals, and you want to highlight the historical evolution of virtualization. Create a timeline that traces the major milestones and innovations in virtualization, starting from its inception to the present day.	3	1	C	P
3	Compare and contrast full virtualization with para-virtualization. Explain their respective advantages and limitations.	3	1	U	C
4	Discuss the major RAID levels and their use cases. How does RAID improve data storage reliability and performance.	3	2	R	F
5	Assume you oversee a website that has significant demand swings, such as a system for selling event tickets. How would you organize the systems transition from a conventional to a cloud environment?	3	2	A	M

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Virtual servers run on a hypervisor, and hardware resources are emulated for the virtual servers via the hypervisors. If the hypervisor fails or if the underlying physical server fails (thereby causing the hypervisor to fail), the failure condition cascades to all of its hosted virtual servers. Design a cloud architecture to overcome this failure.	5	1	An	M
7	In a scenario where a data center needs to reduce its energy consumption and space requirements, what advantages does virtualization offer over a traditional infrastructure?	5	3	U	P
8	A school having nice computer lab facility wants to go on cloud to decrease the cost of licensing software. The infrastructure that the school is having is updated and is in good condition. However, they are reluctant to move on cloud, thinking it to be expensive. Neither they want their database to move out of premises. Examine the requirement of the school and identify the deployment model and cloud services. Also, justify your answer.	5	3	An	M

Course Name: BCSE0551: INTRODUCTION TO BUSINESS ANALYTICS

Course Outcome

CO1: Understand the use of reference architecture in business analytics.
CO2: Explain the decision support science and data warehouse.
CO3: Understand the concepts of Business Intelligence.
CO4: Explain the KDD in data mining.
CO5: Understand the process of analysis using Dashboards and Reports.
CO6: Explain Big Data, Filtering Big Data and Adoption Architecture.

Printed Pages:03 **University Roll No. .....**

Mid Term Examination, Odd Semester 2023-24
B.Tech. (CSE - DA), II, III
BCSE0551: INTRODUCTION TO BUSINESS ANALYTICS

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:

1. Attempt all questions
2. do not switch between section A, and B; first, solve one section, then move to second
3. Use only a blue pen to answer
4. No B sheet will be given

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider you are working for Analytical firm, as a new joinee you suppose to assist the design the data warehouse for firm. They got a project on ecommerce website. You are supposed to design the multi-tier architecture for data warehouse of this client. Draw and explain the multi-tier architecture of data warehouse.	3	1	A	P

2	Rohan had created a list of price of products such as `[23, 45, 67, 88]`
	Help Rohan by writing code to do the following and also show the output:
	a. Create list `price` and print it.
	b. Add `93` and `44` into list `price`. Print the updated list `price`.
	c. Explain Rohan the difference between function append and extend in lists.
	d. Delete the element `45` from the list. Print the updated list.
	e. Delete the element on position (index) 4 from the list. Print the updated list.
	f. What will be the output of following code:
	`price = [23, 45, 67, 88]`
	`price2 = price`
	`price[2] = 67`
		3	2	A	M

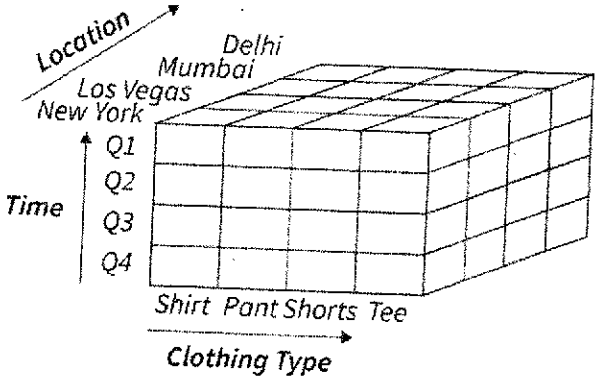
	`print(price2)3`				
3	Shobha is new to your analytical firm. She got the job of collecting data from various sources such as mobile apps, website and company operational website. She getting the data from various sources in various formats. Explain the process she needs to perform before loading the data to data warehouse.	3	2	C	P
4	Shubham recently learns python explain to him how to do the following: a. He created a variable `my_data`, but forgets its datatype and value. Write code to find both these for him. b. He finds he mistakenly add data to `my_data` as `"678"`. Now help him to covert it into appropriate datatype. c. He had a variable `my_str ="Ram Singh is the employee of Amul dairy"`, wants to starting index of `"Amul"`. d. Extract word `"Amul"` from `my_str`, store it in `my_str2` and then reverse it. [Note: extract using some function not from counting index] e. Convert `my_str` to uppercase and print it. f. Extract each word form `my_str` and store in list `my_list`. Print this `my_list`.	3	2	An	M
5	Differentiate between OLAP and OLTP.	3	3	U	C

Section – B

*Attempt **All** Questions*

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Discuss the various classification of Analytics with some examples and case studies?	5	1	U	P
7	Case Study: • Brijwasi Foods & Beverages is a new company which produces dairy, bread and sweet products with production unit located at Mathura. • These products are sold in North, North West and Western region of India. • They have sales units at Mumbai, Pune, Ahmadabad, Delhi and Mathura.	5	3	C	M

	• Address of each above sales unit. • Date and time of sale. • The President of the company wants sales information in terms of unit sold, average sold, and rupees sold. Create star, snowflake and fact constellation for this data. You can add new dimensions and fact as per your design requirement.				
8	Consider a cube that represents the *annual sales (4 Quarters: Q1, Q2, Q3, Q4)* of various kinds of *clothes (Shirt, Pant, Shorts, Tees)* of a company in 4 *cities (Delhi, Mumbai, Las Vegas, New York)* as shown below:  Draw the cube when apply following: a. Drill down to time dimension to Jan, Feb .. Dec. b. Roll-up operation on the given data cube by combining categorizing the sales based on the countries instead of cities. c. Dice operation to retrieve the sales done by the company in the first half of the year i.e., the sales in the first two quarters. d. Slice the sales done by the company during the first quarter. e. Pivot the slice done in part (d).	5	3	E	M

Course Name: IT Application and Data Security
Course Outcome
CO1-Understand software development methodology and application security
CO2- Explain input validation strategies; protect sensitive data, effective authentication and authorization strategies.
CO3- Understand various attacks like DoS, buffer overflow, web specific and database specific.
CO4- Implement security as a culture to show mistakes that make applications vulnerable to attacks
CO5- Demonstrate the security enhancements of web based applications.
CO6- Understand the basics of data security threats.
CO7- Apply the data security threat techniques and its counter measures

Printed Pages:3 **University Roll No. .....**

Mid Term Examination, Odd Semester 2023-24
Program: B. TECH(CSF), Year: II, Semester: III
Subject Code & Name- IT Application and Data Security (BCSE 0601)

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Differentiate between threat and malware. Suppose you are leading a project to develop a new web application. Together with your project team you reprioritize the product backlog, determine velocity for the past iteration, and adapt your work plans accordingly. Additionally, you facilitate retrospectives every two weeks. Identify the suitable development model for complete the project. Also discuss advantage and limitation.	3	1	R	F
2	Discuss any two methods that can be used by attacker to exploited of a input validation vulnerability. Write down the concrete solutions for input validation and discuss the most effective safeguards against it.	3	2	U	C
3	What do you understand by Network eavesdropping? How Awareness, Encryption and Network Segmentation are useful to prevent network eavesdropping vulnerability.	1+2	3	An	F
4	Write short note on the followings: a. session hijacking b. Man in Middle attack c. Configuration management	1+1+1	3	R	F
5	Alexander Rocco Corporation has multiple OSs running in its many offices. Before conducting a	1+1+1	2	A	C

	security test to determine the vulnerabilities you need to correct, you want to determine whether any OSs are running that you're not aware of. The network administrator is resistant to giving you information after learning you are working to discover network security vulnerabilities. The administrator sees you as a threat. After several hours of interviews, you learn only that the network administrator's personal email address is vader2601@gmail.com, and an old Red Hat Enterprise Linux (RHEL) server is running on one of the company's systems. Based on this information, answer the following questions: 1. What tools might you use after learning the network administrator's email address? 2. What can you determine by entering the network administrator's email address into Google? What about entering only the handle vader2601? 3. Could the information you learned from Google be used to conduct vulnerability testing?				

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	**a.** Given e=7 and n=187, encrypt and decrypt the message "GLA" by using the RSA algorithm. Use encoding of 00 to 25 for letters A to Z respectively. **b.** Perform signature generation and verification of the hash value of plaintext 15 by using the values of prime number as 19, primitive root as 10, private key and random integer as 7 respectively using Elgamal Digital Signature Scheme.	2+3	4	U	F
7	Broken authentication is a common security vulnerability that occurs when authentication and session management mechanisms are improperly implemented, allowing attackers to gain unauthorized access to user accounts or administrative functionalities. **a.** Discuss What are the attacks carried out in	3+2	3	E	P

	authentication and Authorization. b. What are the various methods and best practices used to prevent and mitigate broken authentication.				
8	Discuss and explain the countermeasure for these three types of attack scenarios. a. A database application stores passwords of customers without salting the hashes. An attacker is able to access the password file due to a flaw during file upload and could be able to read all the unsalted hashes via rainbow table. b. A web application uses automatic database encryption algorithm to encrypt sensitive data, the credit card numbers of customers, and stores it on the database. The same web application also allows automatic decryption and retrieval of data from the database, thereby permitting an attacker to use SQL injection attack to disclose in clear text the sensitive information, which needs to be protected. c. A web application is not using secure socket layer to authenticate its web pages. This application security vulnerability can be exploited by an attacker who monitors the network traffic and can thieve user's session cookies. The cookie can then be misused to hijack the user's session using the cookie replay attack, thereby allowing the sensitive data of the user to get compromised.	1+2+2	2	U	C

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Mid-Term Examination, Odd Semester 2023-24

B.Tech. CSE, II: Year, III: Semester

BCSC 0081: Blockchain Essentials

Time: 1-Hour Maximum Marks: 30

Note: Attempt all questions. 60 x 0.5 = 30 Marks

1. In Bitcoin, what is the role of a "block header" in the blockchain structure?
a) It contains the transaction data.
b) It links the current block to the previous block.
c) It stores the public keys of miners.
d) It contains the root hash of the Merkle tree.

2. What is the primary purpose of the "nonce" in a Bitcoin block's header?
a) To encrypt the block's transactions
b) To link the block to the previous block
c) To determine the mining difficulty
d) To add randomness to the block's hash

3. In the Bitcoin blockchain, what is the significance of the "genesis block"?
a) It's the first block in the blockchain.
b) It's a special block containing only coinbase transactions.
c) It's the block with the highest proof-of-work.
d) It's a block created by Satoshi Nakamoto.

4. In Bitcoin, what consensus mechanism is used to secure the blockchain and validate transactions?
a) Proof of Stake (PoS)
b) Proof of Work (PoW)
c) Delegated Proof of Stake (DPoS)

d) Practical Byzantine Fault Tolerance (PBFT)

5. How often does a new block get added to the Bitcoin blockchain?
a) Every 10 minutes
b) Every hour
c) Every day
d) Every week

6. Which cryptographic function is primarily used to secure the integrity of transactions in the Bitcoin blockchain?
a) RSA encryption
b) SHA-256 hashing
c) Diffie-Hellman key exchange
d) ECC (Elliptic Curve Cryptography)

7. What is the size limit for a typical Bitcoin block?
a) 1 MB
b) 10 MB
c) 100 MB
d) 1 GB

8. What is the purpose of the "block timestamp" in a Bitcoin block?
a) To indicate the time the block was mined
b) To specify the Bitcoin price at the time of mining
c) To identify the miner's location
d) To set the difficulty level for mining

9. In the Bitcoin blockchain, what is the role of a "public key"?
a) To hide the identity of the sender
b) To encrypt transaction data
c) To verify the authenticity of a digital signature
d) To generate private keys

10. What is the purpose of the "difficulty target" in Bitcoin mining?
a) To determine the miner's location

b) To regulate the rate at which new bitcoins are created
c) To adjust the difficulty of finding a valid block hash
d) To set the maximum block size

11. In the Proof of Work (PoW) consensus mechanism, what is the task that miners must perform to add a new block?
a) Staking a certain amount of cryptocurrency
b) Solving a complex mathematical puzzle
c) Voting for block validators
d) Signing the block with their private key

12. How does the Bitcoin network adjust the difficulty of the PoW puzzle to ensure that new blocks are added approximately every 10 minutes?
a) Miners manually adjust the difficulty.
b) It adjusts automatically based on the number of miners.
c) The difficulty remains constant.
d) It depends on the Bitcoin price.

13. What is one advantage of the Proof of Stake (PoS) consensus mechanism over Proof of Work (PoW) in terms of environmental impact?
a) PoS is more energy-efficient.
b) PoS requires more computational power.
c) PoS has a shorter block confirmation time.
d) PoS doesn't rely on miners.

14. What is the primary challenge or drawback of the Proof of Stake (PoS) consensus mechanism?
a) It is susceptible to Sybil attacks.
b) It consumes a significant amount of energy.
c) It requires a high level of computational power.
d) It can lead to centralization.

15. In PoW, what is the task that miners must perform to add a new block to the blockchain?
a) Solve a complex mathematical puzzle

b) Vote for the next block
c) Sign the previous block
d) Verify all transactions in the block

16. What is the purpose of the "nonce" in the PoW mining process?
a) To identify the miner's identity
b) To add randomness to the block's hash
c) To create new bitcoins
d) To encrypt transaction data

17. In the PoW process, what happens if a miner successfully solves the puzzle and adds a new block to the blockchain?
a) They receive a reward of newly created bitcoins.
b) They become a full node on the network.
c) They gain voting rights in the network.
d) They automatically become a miner in the next round.

18. What is a "51% attack" in the context of PoW?
a) A miner controlling more than 51% of the network's computational power
b) A miner having 51% of the bitcoins in circulation
c) A miner discovering 51% of all transactions
d) A miner reaching 51% of all mining rewards

19. Which type of blockchain fork is considered backward-compatible and does not necessarily result in the creation of a new cryptocurrency?
a) Soft fork
b) Hard fork
c) Segregated fork
d) Layered fork

20. In blockchain what happens to the transactions and balances of users who do not upgrade their software?
a) Their transactions are reversed.
b) Their balances are unaffected.

c) Their balances are doubled.
d) Their transactions become invalid.

21. What is the name of the most well-known hard fork of Bitcoin that resulted in the creation of a new cryptocurrency with larger block sizes?
a) Bitcoin Classic
b) Bitcoin Unlimited
c) Bitcoin Cash
d) Bitcoin Gold

22. In a Bitcoin transaction, what is the "transaction fee" used for?
a) To pay taxes on the transaction
b) To reward miners for including the transaction in a block
c) To verify the recipient's identity
d) To encrypt the transaction data

23. What is the primary purpose of a cryptographic hash function?
a) Encryption
b) Data compression
c) Data integrity verification
d) Key generation

24. Which hash algorithm is commonly used for secure password storage and authentication?
a) MD5
b) SHA-1
c) SHA-256
d) CRC32

25. Which of the following is a potential security vulnerability of hash functions?
a) Key exposure
b) Brute-force attacks
c) Birthday attacks
d) Data compression

26. Which cryptographic hash algorithm is considered broken and unsuitable for most security applications due to vulnerabilities?
a) SHA-256
b) SHA-1
c) MD5
d) HMAC

27. What is the typical length, in bits, of the hash output generated by the SHA-256 algorithm?
a) 4 bytes
b) 16 bytes
c) 32 bytes
d) 256 bytes

28. Which hash function is commonly used in the blockchain technology for creating a secure link between blocks?
a) MD5
b) SHA-256
c) HMAC
d) CRC32

29. Which of the following hash functions is most resistant to length extension attacks?
a) MD5
b) SHA-1
c) SHA-256
d) HMAC

30. In the context of hash functions, what does it mean for a hash to have a "birthday bound"?
a) The hash has a long lifespan.
b) The hash has a fixed output size.
c) The probability of a collision exceeds 50%.
d) The hash is resistant to preimage attacks.

31. In a blockchain, why is it important to use a secure cryptographic hash function for linking blocks?
a) To reduce computational overhead
b) To ensure faster block validation
c) To prevent the manipulation of previous block data
d) To improve transaction throughput

32. In RSA encryption, which mathematical operation is computationally intensive and forms the basis of the algorithm's security?
a) Division
b) Prime Factorisation
c) Exponentiation
d) Addition

33. What is the purpose of the Euler's totient function (φ) in RSA encryption?
a) To generate the public key
b) To calculate the private key
c) To determine the message length
d) To find coprime integers for key generation

34. In RSA encryption, if two parties want to communicate securely, which key is used for encryption?
a) Sender's private key
b) Receiver's public key
c) Receiver's private key
d) Sender's public key

35. Which of the following is NOT a requirement for RSA key generation?
a) Two large prime numbers
b) Euler's totient function value
c) Modular exponentiation
d) Public exponent

36.In RSA encryption, why is it important to choose large prime numbers for key generation?
a) To increase the size of the ciphertext
b) To reduce computational overhead
c) To enhance the security of the encryption
d) To decrease the encryption and decryption time

37. Which attack on RSA encryption involves computing the private key from the knowledge of the public key and a set of ciphertexts?
a) Factoring attack
b) Brute-force attack
c) Chosen plaintext attack
d) Known plaintext attack

38. What does RSA stand for in the context of cryptography?
a) Random Secure Algorithm
b) Rapid Symmetric Algorithm
c) Rivest-Shamir-Adleman
d) Reliable Secret Authentication

39. What is the typical purpose of RSA encryption in secure communication?
a) Data compression
b) Data integrity verification
c) Confidentiality
d) Digital signatures

40. In RSA, what is the mathematical relationship between the public and private keys?
a) They are identical.
b) The public key is derived from the private key.
c) The private key is derived from the public key.
d) There is no mathematical relationship between them.

41. What is the primary purpose of the public key in RSA encryption?
a) Encrypting data
b) Decrypting data
c) Signing data
d) Verifying signatures

42. In RSA, if someone encrypts a message using your public key, who can decrypt it?
a) Only the person who encrypted it
b) Only you with your private key
c) Anyone with access to the public key
d) No one; it cannot be decrypted

43. What is the primary advantage of RSA encryption over symmetric encryption algorithms like AES?
a) Faster encryption and decryption
b) Simplicity of key management
c) Ability to securely share keys
d) Asymmetric key distribution

44. What does ECC stand for in the context of cryptography?
a) Elliptic Curve Calculator
b) Elliptical Cryptographic Cipher
c) Elliptic Curve Cryptography
d) Encrypted Cipher Control

45. Which mathematical structure forms the basis of ECC encryption?
a) Prime numbers
b) Elliptic curves
c) RSA modulus
d) Fibonacci sequences

46. Compared to traditional public-key encryption like RSA, ECC typically requires:
a) Larger key sizes for equivalent security

b) Smaller key sizes for equivalent security
c) The same key sizes for equivalent security
d) More computational resources

47. In ECC, what is a public key represented as on an elliptic curve?
a) A point on the curve
b) A scalar value
c) A prime number
d) A matrix

48. Which ECC-based encryption algorithm is commonly used for secure communications over the internet, including HTTPS connections?
a) ECDSA (Elliptic Curve Digital Signature Algorithm)
b) ECDH (Elliptic Curve Diffie-Hellman)
c) ECIES (Elliptic Curve Integrated Encryption Scheme)
d) ECCC (Elliptic Curve Cryptographic Cipher)

49. In ECC, what is the ECC domain parameter?
a) A fixed point on the curve
b) A curve equation
c) A user's public key
d) A secret key

50. What is the primary purpose of a digital signature in cryptography?
a) Data encryption
b) Data compression
c) Data integrity verification
d) Data obfuscation

51. Which cryptographic property ensures that a digital signature cannot be forged or altered without detection?
a) Confidentiality
b) Integrity
c) Authentication
d) Non-repudiation

52. How does a recipient verify the authenticity and integrity of a digitally signed message?
a) By comparing the message to a pre-shared secret key
b) By decrypting the message with the sender's public key
c) By re-computing the hash of the message and comparing it to the signature
d) By contacting a certificate authority (CA) for verification

53. What is the purpose of a timestamp in a digital signature scheme?
a) To prove the age of the digital signature
b) To increase the security of the private key
c) To encrypt the digital signature
d) To add a watermark to the signed document

54. In symmetric-key cryptography, what do both the sender and receiver use to encrypt and decrypt messages?
a) Different keys
b) Public keys
c) Private keys
d) The same key

55. What is blockchain halving?
a) A process of reducing the number of miners in the blockchain network
b) A security measure to prevent double-spending attacks
c) A protocol upgrade to increase block sizes
d) A predetermined reduction in block rewards at specific intervals

56. What happens to the block rewards for miners during a blockchain halving event?
a) They double in value
b) They remain the same
c) They are reduced by half
d) They increase by 25%

57. What is the approximate block reward for miners in Bitcoin after the most recent halving event?

a) 12.5 BTC
b) 25 BTC
c) 6.25 BTC
d) 50 BTC

58. Which of the following is NOT a common solution proposed for improving blockchain scalability?

a) Sharding
b) Off-chain scaling solutions
c) Increasing the block size limit
d) Reducing security measures

59. What is the Lightning Network, and how does it contribute to blockchain scalability?

a) A blockchain security protocol
b) A second layer scaling solution for Bitcoin
c) A type of blockchain mining algorithm
d) A hardware device for mining

60. What trade-offs may be associated with implementing blockchain scalability solutions?

a) Increased decentralization and higher security
b) Reduced security and increased complexity
c) Slower transaction processing and lower efficiency
d) Reduced transparency and lower costs

Course Name: Discrete Mathematics

Course Outcome

CO1: Understand the notion of mathematical thinking and proofs to solve the problem.
CO2: Apply the basics of discrete probability and number theory to solve the real world problem.
CO3: Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5: Analyze the validity of an argument using logical notation.
CO6: Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7: Understand the basic principles of sets, set equalities and operations in sets.
CO8: Apply counting principles to determine probabilities.

Printed Pages: 02 **University Roll No.** .....

Mid Term Examination, Odd Semester 2023-24
B. TECH(CS/AIML), II/III Year
BCSC 1010 DISCRETE MATHEMATICS

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	A computer company receives 350 applications from computer graduates for a job planning a line of new Web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?	3	CO4	E	P
2	How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen? How many must be selected to guarantee that at least three hearts are selected?	3	CO8	An	C
3	Show that the relation R on the set of all bit strings such that s R t if and only if s and t contain the same number of 1s is an equivalence relation.	3	CO7	E	P
4	(a) Suppose that there are eight runners in a race. The winner receives a gold medal, the second place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award these medals, if all possible outcomes of the race can occur and there are no ties? (b) How many different 11 letter words are possible using the letters of the word COMBINATION such that 2 O's come together ?	3	CO8	An	C

5	Conjecture a formula for the sum of the first n positive odd integers. Then prove your conjecture using mathematical induction.	3	CO1	C	C

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Solve the given equation using generating function $a_n = 8a_{n-1} + 10^{n-1}$, $a_2 = 82$	5	CO8	E	P
7	Answer these questions for the poset ({3, 5, 9, 15, 24, 45}, |). a) Find the maximal elements. b) Find the minimal elements. c) Is there a greatest element? d) Is there a least element? e) Find all upper bounds of {3, 5}.	5	CO1	E	P
8	A popular puzzle of the late nineteenth century invented by the French mathematician Édouard Lucas, called the Tower of Hanoi, consists of three pegs mounted on a board together with disks of different sizes. Initially these disks are placed on the first peg in order of size, with the largest on the bottom. The rules of the puzzle allow disks to be moved one at a time from one peg to another as long as a disk is never placed on top of a smaller disk. The goal of the puzzle is to have all the disks on the second peg in order of size, with the largest on the bottom. Find the recurrence relation and solve it.	5	CO8	A	C

Course Name: DATABASE MANAGEMENT SYSTEM

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real life application, with constraints and keys using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

Printed Pages: 03 **University Roll No. .....**

Mid Term Examination, Odd Semester 2023-24
B. TECH. (CS), Year: II, Semester: III
Subject Code: BCSC 1003
Subject Name: Database Management System

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Mention clearly if any assumptions are being made.
2. No 'B' sheet will be given.

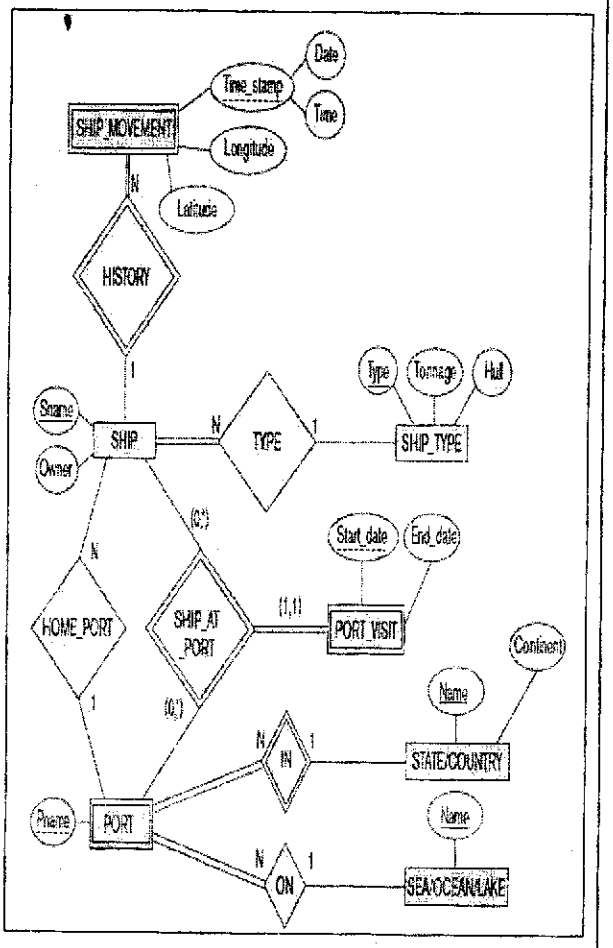
Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	*"DBMS is preferred by industries over File Systems".* Justify your answer with proper reasons and scenario.	3	1	U	C
2	Assume that you are appointed in your dream company as a database designer and got a project to work on backend implementation. Your job profile includes designing a complete database for a web based project. Identify and briefly explain all the phases you need to handle during the database development life cycle during this project.	3	1	U	C
3	Write short note on: (a) Hierarchical Data Model, (b) Network Data Model, and (c) Relational Data Model.	3	2	R	F
4	Define RENAME (ρ) operator in details. Is it possible to apply SELECT (σ) operator and PROJECT (π) operator in a single SQL syntax? If yes, then explain how. If no, then explain why.	3	3 & 4	An	C
5	Consider a relation schema R = (A, B, C, D, E, F) on which the following functional dependencies hold: {A → B, BC →D, E → C, D → A} Identify the Candidate Key(s) of R.	3	4	E	P

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider the ER diagram given below & convert it into all possible tables: 	5	3	An	P
7	Suppose following tables are created in a database: College (cName, state, enrollment) Student (sID, sName, GPA, sizeHS) Apply (sID, cName, major)	5	5	C	P

	Now, write the SQL syntax for the following queries: a. Add cName as primary key in the College table. b. Make cName in Apply table as foreign key referring college table. c. Rename column state to address in the College table. d. Increase size of address attribute in College table from 15 to 25. e. Drop primary key from the College table.				
8	Consider that a relation STUDENT_PERFORMANCE (Name, CourseNo, RollNo, Grade) has the following functional dependencies: {Name, CourseNo → Grade, RollNo, CourseNo → Grade, Name → RollNo, RollNo → Name} Determine the highest normal form of this relation.	5	6	E	C

Course Name: **Software Engineering**

Course Outcome

- CO1: Understand the basic concepts of software engineering.
- CO2: Apply software processes to solve real world problems.
- CO3: Estimate the cost, effort and schedule of software using COCOMO Model.
- CO4: Analyze the software design techniques (structure chart, SDM, sequence diagram).
- CO5: Understand the basic concepts of OO analysis and design.
- CO6: Develop the test cases to validate the software.
- CO7: Understand the basic models of software Quality and maintenance.

Printed Pages: 3 **University Roll No. .....**

Mid Term Examination, Odd Semester 2023-24
B.Tech (CSE)/B.Tech (EC) 2nd Year/3rd Year,
BCSC0009 – Software Engineering

Time: 2 Hours **Maximum Marks: 30**

Attempt All Questions:-

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Compare the life cycle models based on their distinguishing factors, strengths and weaknesses.	3	1	U	F
2	Describe the characteristics of software contrasting it with the characteristics of hardware. Illustrate with a diagram that the software does not wear out.	3	2	R	F
3	Assume that you want to design software for University Registration System Management with common facilities such as. 1. Professors indicate which courses they will teach on-line. 2. A course catalog can be printed 3. Allow students to select on-line four courses for upcoming semester. 4. No course may have more than 10 students or less than 3 students. 5. When the registration is completed, the system sends information to the billing system. 6. Professors can obtain course rosters on-line. 7. Students can add or drop classes on-line. Draw a use case diagram for this software.	3	2	A	P
4	Identify at least two functional & one non – functional requirement of a Bank Automated Teller Machine	3	1	A	P

	(ATM) system. Enlist various characteristics of an SRS.				
5	A).A software project is estimated to have a size of 50,000 lines of code and an effort of 200 person-months, using the basic COCOMO model. If the project is expected to be completed in 15 months, what is the required team size in terms of the number of people? B).A software project is estimated to have a size of 25,000 lines of code, using the intermediate COCOMO model. The project complexity is rated as "nominal". If the estimated effort is 150 person-months, what is the productivity rate of the development team in lines of code per person-month?	3	3	A	P

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Using the example of a retail clothing store in a mall, list relevant data flows, data stores, processes, and sources/sinks. Observe several sales transactions. Draw a context diagram and a level-1 diagram that represent the selling system at the store. Explain why you chose certain elements as processes versus sources/sinks.	5	2	An	P
7	An application has the following: 10 low external inputs, 12 high external outputs, 20 low internal logical files, 15 high external interface files, 12 average external inquiries. In addition to this, the system requires significant data communication, and other complexity adjustment factors are treated as	5	3	A	P

	moderate. Compute the Unadjusted function point and Function Point of the project The software is going to be implemented in C language so also calculate the size of the project.				
8	You have been hired as a project manager to lead the development of a new web-based application for a small e-commerce company. Together with your project team you reprioritize the product backlog, determine velocity for the past iterations, and adapt your work plans accordingly. Additionally, you facilitate retrospectives every two weeks. The company has a tight deadline and limited resources, and they need the application to be completed within one or two months. What specific methodology you would use and why? Explain the process with its pros and cons and neat sketch.	5	3	An	P

Functional Units	Weighing factors		
	Low	Average	High
External Inputs (Ei)	3	4	6
External Output (EO)	4	5	7
External Inquiries (EQ)	3	4	6
External logical files (ILF)	7	10	15
External Interface files (EIF)	5	7	10

Project	a_i	b_i	c_i	d_i
Organic	3.2	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	2.8	1.20	2.5	0.32

Software Project	a_b	b_b	c_b	d_b
Organic	2.4	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32